

Master Thesis Proposal

Rewiring the market clock in Spanish electricity markets

Pablo Paramio Pérez
Advisor: Pedro Mira

CEMFI

February 24, 2026

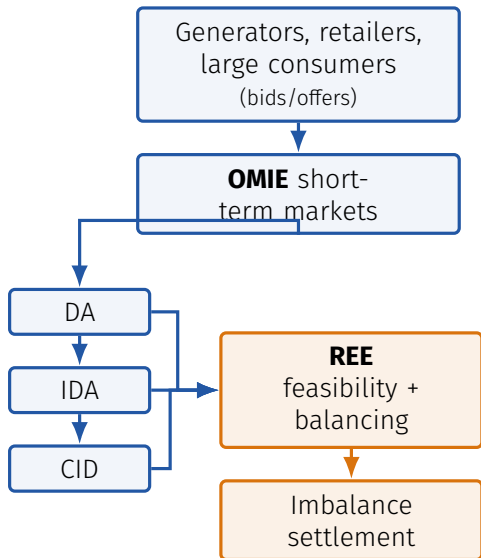
Some motivation and context

Motivation

- **Having electricity in this room right now is now trivial at all:** it is an homogeneous good that must be delivered continuously from many different technologies in many different places.
- Therefore, at every moment, the system must solve a difficult coordination problem:
 - many generators with different costs and technical constraints,
 - many consumers/retailers with changing demand,
 - network limits and operational security requirements,
 - constant forecast updates (demand, wind, solar, outages).
- How do we price electricity? ⇒ **Markets are the coordination device:**
 - they aggregate dispersed information through bids/offers,
 - produce prices and schedules for delivery,
 - and allow participants to revise positions as uncertainty resolves.
- But electricity is not a standard commodity: market outcomes must remain compatible with **real-time system security**, so a **system operator** and **balancing mechanisms** are still essential.

Context: Spanish electricity system

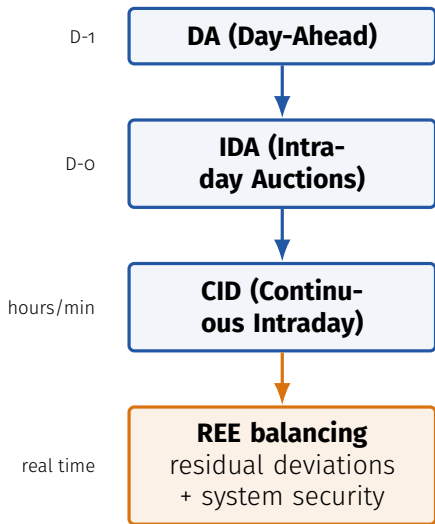
- Electricity trading and physical operation must be coordinated tightly.
- For this presentation, I separate the short-term system into two layers:
 - OMIE^a (markets):** Day-Ahead, Intraday Auctions, Continuous Intraday.
 - REE^b (system operator / TSO):** feasibility checks, balancing activation, secure operation.
- Market outcomes are **schedules/plans**; REE ensures they are **physically deliverable**.
- Remaining deviations are handled through **balancing** and **imbalance settlement**.



^aEnergy Iberian Market Operator

^bSpanish Electricity Network

What happens from D-1 to delivery? DA, IDA, CID and balancing



- **DA (Day-Ahead):** the main market stage. Participants submit bids/offers for next-day delivery; the outcome is a **baseline schedule** and market prices. Uniform price auction.
- **IDA (Intraday Auctions):** several **discrete sessions** to update positions as forecasts and availability change. Uniform price for each one.
- **CID (Continuous Intraday):** **continuous matching** close to delivery; provides the last voluntary adjustment opportunity. Different price for each transaction
- **Balancing and Imbalance settlement (REE):** REE manages remaining system deviations (and security constraints) and deviations from final schedules are settled financially.

2024–2025 reform sequence

- Traditionally, the market time unit (MTU) of the auctions was hourly (MTU60).
- The clock rewiring happened in stages:
 - Dec 2024:** ISP15 (imbalance settlement becomes 15-min)
 - Mar 2025:** ID15 (intraday products become 15-min)
 - Oct 2025:** DA15 (day-ahead products become 15-min)
- This creates transitional regimes where **settlement** and **trading** granularities differ.

Period	ISP	DA	ID
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	60	15
Oct25 onward	15	15	15

Minutes (60 = hourly, 15 = quarter-hourly).



Research question

Research question

This thesis will **empirically assess the impact of the MTU reform** and will extend the theoretical framework in my previous research proposal to better capture firms' strategic behaviour in different MTU situations. The thesis will (potentially) have the following building blocks:

- **Reduced-form analysis (main):** implement a reduced-form evaluation of the reform with all the available data, using different outcomes of interest (slide 8).
- **Theoretical model and structural modelling:** extend [Ito and Reguant \(2016\)](#) sequential market model to better capture MTU strategic differences^a and build a structural auction model to study some economic mechanisms (slide 12).

^aPartially done in my Energy economics research proposal.

Theoretical model

Theoretical model

- In my energy research proposal, I already built an extension of [Ito and Reguant \(2016\)](#) with the following conclusions, both under full and strategic arbitrage across periods:
 - For efficiency, DA15 is the key reform that smooths imbalances.
 - Dispersion risk across quarters is concentrated in the transitional period DA60/ID15.
- However, my model drew important conclusions only for the transitional situation, not for a lower MTU itself (beyond the obvious efficiency improvement).
- MTU15 creates new opportunities for sophisticated agents and technologies, so extending the model to a setting with winners and losers could be very interesting.
- Also, a complete welfare analysis was absent from my model, so I could extend it in that direction as well.
- Nevertheless, **the main purpose of this project is to assess the reform empirically.**

Data and Empirical strategy

Data

- **Core market data (OMIE):** publicly available day-ahead and intraday prices and quantities (*last six rolling years*), including intraday auction outcomes, continuous intraday transactions, and accepted bid-curve information.
- **Key dataset for firm behavior:** disaggregated unit-level bids (*Detalle de las ofertas*), with a **3-month publication lag** (currently available up to **Nov 2025**). It contains bid price-quantity pairs by unit and period, allowing the reconstruction of granular bidding behavior (by firm/technology, using anonymized identifiers).
- **Controls and mechanisms:** weather data from AEMET; balancing outcomes, system-operation costs, demand, generation mix, storage, and interconnection flows from REE/ESIOS (via API).

Current status

I am building a fully automated data pipeline for OMIE data. The collection architecture and DA price automation have already been implemented. This is crucial because the 3-month lag limits current post-DA15 (Oct 2025) coverage.

Outcomes of interest

- **Arbitrage price wedge (sequential efficiency):** DA vs IDA for the same delivery period.

$$W_{d,h}^{DA/IDA} = P_{d,h}^{DA} - \bar{P}_{d,h}^{ID}, \quad \bar{P}_{d,h}^{ID} = \sum_{j \in ID} \frac{\text{vol}_{d,h,j}}{\sum_{j \in ID} \text{vol}_{d,h,j}} P_{d,h}^{ID(j)}.$$

- **Continuous-market VWAP near delivery:** for window W (e.g. last 2 hours),

$$\hat{P}_{d,h}^{CIM}(W) = \sum_{\ell \in \mathcal{L}_{d,h}(W)} p_{\ell} \frac{Q_{\ell}}{\sum_{k \in \mathcal{L}_{d,h}(W)} Q_k}, \quad W^{DA/CIM} = P_{d,h}^{DA} - \hat{P}_{d,h}^{CIM}(W).$$

- For the DA60-ID15 scenario, we could compare DA vs each quarter or with the average, both for the IDA and the continuous market.
- **Liquidity / thinness (CID):** #trades, total volume, concentration; time-to-delivery profiles (fragmentation vs depth).
- **Schedule revision intensity:** physical gap between the initial market plan and the final delivery schedule (REE adjustments).

Empirical strategy: why not DiD?

- **No clean external counterfactual:** most comparable European markets adopted MTU15 and differ structurally (technologies, interconnections, institutions).
- The evaluation therefore relies on **before/after** designs with:
 - rich fundamentals (load, wind/solar, temperature, fuel/CO₂, outages),
 - seasonality and calendar structure,
 - and explicit introduction of the reform (Dec 2024 / Mar 2025 / Oct 2025).

Alternative

Another novel strategy, used in [Märkle-Huß et al. \(2018\)](#) for the European Power Exchange (EPEX) MTU reform (2011 and 2014), is to apply a **Bayesian structural time series approach**, as proposed by [Brodersen et al. \(2015\)](#). The idea is to construct a data-driven counterfactual of the outcome variable using only pre-intervention information.

Brodersen et al. (2015): main idea

For each outcome series (e.g. wedge, dispersion, liquidity, balancing metric):

- **Step 1 (train pre):** fit a state-space model on the pre-reform period with:
 - local trend + seasonality,
 - fundamentals as covariates (AEMET/REE/ESIOS, fuels/CO₂),
 - spike-and-slab prior to select predictors (avoids overfitting).
- **Step 2 (predict post):** generate the posterior predictive distribution of the no-reform path.
- **Step 3 (impact):** the effect is observed minus counterfactual, with credible intervals and cumulative impact.

What pre-reform periods should we consider?

It is far from trivial. The most important reforms for **wholesale** market prices are ID15 and DA15, ISP15 only affects directly security/balancing reserve markets, so for ID15 we can consider all the previous available years. However, it depends on the importance of balancing on the outcome of interest.

What about the April 28 blackout?

- As expected, the blackout is a huge disturbance to assess the effect of MTU15.
- It affects both the **post-ID15 reform** period and the **pre-DA15 reform** period.
 - For ID15, I propose to focus on the post-reform period before the blackout (March 19th - April 28th). There were also a couple of days with imbalances (April 22nd), so maybe we should also drop them too for ID15 evaluation.
 - For DA15, I propose to use the blackout-reform period (April 28th-October 1st) and the post-reform.
- Moreover, note that **REE has increased the security requirements and balancing reserves** for precaution, so the impact assessment of DA15 has to be evaluated in this context.

Structural model (?)

Brief description

- This is the **most ambitious part of the project**. With a 3-month lag, we have **very rich disaggregated bidding data** (*Detalle de las ofertas*), which include bids, technology, and plant ownership.
- With a structural model, we could answer questions about:
 - **Thinness / residual-demand mechanism:** do firms face steeper residual demand in specific quarters (especially near delivery) after MTU15?
 - **Strategic bidding response:** do firms change the shape or steepness of bid schedules when products become finer?
 - **Heterogeneity:** are responses stronger for flexible technologies (hydro, CCGT) or for pivotal firms?
 - **DA vs ID allocation:** does strategic activity shift from day-ahead commitments to intraday adjustment?
- A very similar exercise is conducted in [Hortaçsu et al. \(2017\)](#), who study whether more sophisticated firms in the Texas electricity market deviate more from Nash equilibrium bidding, finding that increases in sophistication improve system efficiency. A general reference for this type of model is [Gentry et al. \(2018\)](#), although multi-unit uniform-price auctions are not covered in detail.

Thank you!

References

References I

- BRODERSEN, K. H., F. GALLUSSER, J. KOEHLER, N. REMY, AND S. L. SCOTT (2015): “Inferring causal impact using Bayesian structural time-series models,” *The Annals of Applied Statistics*, 9.
- GENTRY, M. L., T. P. HUBBARD, D. NEKIPELOV, AND H. J. PAARSCH (2018): “Structural Econometrics of Auctions: A Review,” *Foundations and Trends(R) in Econometrics*, 9, 79–302.
- HORTAÇSU, A., F. LUCO, S. L. PULLER, AND D. ZHU (2017): “Does Strategic Ability Affect Efficiency? Evidence from Electricity Markets,” Working Paper 23526, National Bureau of Economic Research.
- ITO, K. AND M. REGUANT (2016): “Sequential Markets, Market Power, and Arbitrage,” 106, 1921–1957.
- MÄRKLE-HUSS, J., S. FEUERRIEGEL, AND D. NEUMANN (2018): “Contract durations in the electricity market: Causal impact of 15min trading on the EPEX SPOT market,” 69, 367–378.